

Midterm

● Graded

Student

Qiulin Fan

Total Points

90.5 / 100 pts

Question 1

Question1

10.5 / 12 pts

1.1 Part (a)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

1.2 Part (b)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

1.3 Part (c)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

1.4 Part (d)

■ 1.5 / 3 pts

+ 3 pts Correct

✓ + 0 pts Partially correct (see comments)

💬 + 1.5 pts Point adjustment

4

should be additive identity

Question 2

Question 2

15 / 15 pts

2.1 Part (a)

3 / 3 pts

+ 0 pts Incorrect

💬 + 3 pts Point adjustment

2.2 Part (b)

3 / 3 pts

+ 0 pts Incorrect

💬 + 3 pts Point adjustment

2.3 Part (c)

3 / 3 pts

+ 0 pts Correct

💬 + 3 pts Point adjustment

2.4 Part (d)

3 / 3 pts

+ 0 pts Incorrect

💬 + 3 pts Point adjustment

2.5 Part (e)

3 / 3 pts

✓ + 0 pts Incorrect

💬 + 3 pts Point adjustment

Question 3

Question 3

12 / 15 pts

3.1 Part (a)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

3.2 Part (b)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

3.3 Part (c)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

3.4 Part (d)

0 / 3 pts

+ 3 pts Correct

✓ + 0 pts Partially correct (see comments)

3 the subset $\{0\}$ is itself a ring, but not a subring since it doesn't contain 1

3.5 Part (e)

3 / 3 pts

✓ + 3 pts Correct

+ 0 pts Partially correct (see comments)

Question 4

Question 4

15 / 15 pts

4.1 Part (a)

5 / 5 pts

+ 0 pts Incorrect

💬 + 5 pts Point adjustment

4.2 Part (b)

5 / 5 pts

+ 0 pts Incorrect

💬 + 5 pts Point adjustment

4.3 Part (c)

5 / 5 pts

+ 0 pts Incorrect

💬 + 5 pts Point adjustment

Question 5

Question 5

9 / 12 pts

5.1 Part (a)

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Partial credit

- 3 pts Incorrect

5.2 Part (b)

3 / 3 pts

✓ - 0 pts Correct

- 1.5 pts Missing one direction

- 3 pts Partial credit (see adjustments)

- 3 pts Missing

5.3 Part (c)

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Partial credit

- 3 pts Missing

5.4 Part (d)

0 / 3 pts

- 0 pts Correct

✓ - 3 pts Missing

- 3 pts Partial credit

Question 6

Question 6

16 / 16 pts

6.1 Part (a)

4 / 4 pts

✓ - 0 pts Correct

- 2 pts Partial credit

6.2 Part (b)

4 / 4 pts

+ 0 pts Correct

💬 + 4 pts Point adjustment

6.3 Part (c)

4 / 4 pts

✓ - 0 pts Correct

- 4 pts Incorrect

6.4 Part (d)

4 / 4 pts

✓ - 0 pts Correct

Question 7

Question 7

13 / 15 pts

7.1 Part (a)

5 / 5 pts

✓ - 0 pts Correct

- 1.5 pts Insufficient justification

7.2 Part (b)

5 / 5 pts

✓ - 0 pts Correct

- 1 pt Missing proof of K contains both I and J

- 4 pts Missing proof of K is an ideal

- 2 pts Insufficient justification

7.3 Part (c)

3 / 5 pts

- 0 pts Correct

- 5 pts Missing

- 1 pt Insufficient justification

✓ - 5 pts Partial credit (see adjustment)

💬 + 3 pts Point adjustment

1 Need to prove surjectivity as well

2 This should be sending a to $a+J$

Winter 2024

Midterm Exam

Math 412

Name: *Quilin Fan*

Math 412 Winter 2024 Midterm Exam

Time: 120 mins.

1. Answer each question in the space provided. If you require more space, you may use the blank page at the end of this exam, but you must clearly indicate in the provided answer space that you have done so.
2. You may use any results proved in class, on the homework, or in the textbook, except for the specific question being asked. You should clearly state any facts you are using.
3. Remember to show all your work.
4. No calculators, notes, or other outside assistance allowed.

Best of luck!

Problem 1 (12 points). In the following parts, complete the definition of the italicized mathematical term. Be precise and include quantifiers as needed.

- a) Let R be a ring. An ideal $I \subset R$ is *principal* if ...

$$I = \{rs \mid r \in R\} \text{ for some } s \in R$$

- b) Let R be a commutative ring. A non-zero element $r \in R$ is a *zero divisor* if ...

there exists some non-zero element $s \in R$ such that

$$rs = sr = 0_R$$

- c) Let $\mathbb{F}$ be a field and let $\mathbb{F}[x]$ be the polynomial ring with coefficients in $\mathbb{F}$. A non-constant polynomial $p(x) \in \mathbb{F}[x]$ is *irreducible* if ...

it can not be factorized into ~~nonconstant~~ ^{nonconstant} polynomials of lower degree.
 (the only factors of it are ~~constants~~ ^{constants in $\mathbb{F}[x]$} and its associates)

- d) The *kernel* of a ring homomorphism is ...

the set of the elements in the source that are mapped to the multiplicative identity in the target

Problem 2 (15 points). For each of the questions below, give an example with the required properties. **No explanations required.**

- a) A ring with exactly 6 invertible elements.

$$\mathbb{Z}_7$$

- b) A ring with infinitely many elements that has characteristic¹ 3.

$$\mathbb{Z}_3[x]$$

- c) Integers a and b such that $a5 + b13 = 1$.

$$13 = 2 \times 5 + 3$$

$$5 = 3 + 2$$

$$3 = 2 + 1$$

$$2 = 1 \times 2 + 0$$

$$1 = 2 \times 13 - 4 \times 5 - 5 = 2 \times 13 - 5 \times 5$$

$$\underline{a = -5, b = 2}$$

¹The characteristic of a ring R is the positive integer n such that $\underbrace{1_R + \cdots + 1_R}_{n \text{ times}} = 0_R$ or 0 if no such n exists.



Problem 2 Continued... For each of the questions below, give an example with the required properties. No explanations required.

- d) A ring R and an idempotent element $e \in R$ (i.e. $e^2 = e$) where $e \neq 0_R, 1_R$.

$$\underline{[3]_6 \in \mathbb{Z}_6}$$

- e) A polynomial in $\mathbb{Z}[x]$ that is not irreducible in $\mathbb{Z}[x]$ but is irreducible in $\mathbb{Q}[x]$.



Problem 3 (15 points). For each of the questions below, indicate clearly whether the statement is *true* or *false*, and give a short justification or counterexample.

- a) Given any ring R , there exists exactly one ring homomorphism $\mathbb{Z} \rightarrow R$.

True.

$$\varphi(1) = 1_R$$

$$\varphi(n) = \varphi(\underbrace{1 + \dots + 1}_{n \text{ times}}) = n \cdot 1_R$$

So every $n \in \mathbb{Z}$ is mapped to $n \cdot 1_R$, determining the whole homomorphism

- b) Let F is a field, then $F[x]$ is a field.

False.

Because we know the only units in $F[x]$ are the same units in F , and the other elements in $F[x]$ are not unit

For example $\mathbb{R}$ is field but $\mathbb{R}[x]$ is not field,

$x+1$ ~~is~~ has not multiplicative inverse

- c) If R is a commutative ring and $(g) = R$, then g is a unit.

True

Since $(g) = R$, $\exists f \in R$ s.t. $gf = 1_R$

$\Rightarrow g$ is unit.

Problem 3 Continued... For each of the questions below, indicate clearly whether the statement is *true* or *false*, and give a short justification or counterexample.

d) A subset of a ring that is also a ring is a subring.

True.

$S \subseteq R$ is a ring
Since an operation can at most have one identity,
 $1_S = 1_R$ and $0_S = 0_R$

And S is closed under $+$, $\times$ and multiplicative inverse
since it is a ring

So S is a subring of R .

e) If I and J are ideal in a ring R , $I \cup J$ is an ideal in R .

False

Consider $(2), (3)$ are ^{principal} ideals of $\mathbb{Z}$

$$2 \in (2), 3 \in (3)$$

$$\text{but } 2+3=5 \notin (2) \cup (3)$$

Problem 4 (15 points). There are three parts to this problem.

- a) Solve $[10]x + [27] = [0]$ for x in $\mathbb{Z}_{71}$.

$$[10]_{71}x = [-27]_{71}$$

$$[10]_{71}x = [44]_{71}$$

We only need to solve $[10]_{71}y = [1]_{71}$ and then $x = [44]_{71} \cdot y$.

Since $(10, 71) = 1$, $\exists r, s$ s.t. $10r + 71s = 1$

We get $10 \cdot (-7) + 71 \cdot 1 = 1 \Rightarrow y = [-7]_{71} = [64]_{71}$

$$\text{So } x = [44 \times 64]_{71} = [47]_{71}$$

- b) In $\mathbb{Z}_5[x]$ we have that

$$x^3 + 2x^2 + 4x + 1 = x(x^2 + 2x + 1) + (3x + 1)$$

and

$$x^2 + 2x + 1 = 2x(3x + 1) + 1$$

Use this to find the multiplicative inverse of $(x^2 + 2x + 1)$ in $\mathbb{Z}_5/(x^3 + 2x^2 + 4x + 1)$.

By the Euclidean Algorithm for polynomials, (since $\mathbb{Z}_5$ is a field because 5 is prime, the algorithm holds)

(denote $x^2 + 2x + 1$ by $f(x)$,

$x^3 + 2x^2 + 4x + 1$ by $g(x)$)

$$\gcd(g, f) = \gcd(f, 3x + 1) = \gcd(3x + 1, 1) = 1$$

Therefore by reverse Euclidean, $1 = x^2 + 2x + 1 - 2x(3x + 1)$

$$= f - 2x(g - xf)$$

$$= f(2x^2 + 1) - 2x \cdot g$$

$$f(2x^2 + 1) = 1 + 2x \cdot g = 1 \Rightarrow [2x^2 + 1] \text{ is the multiplicative inverse of } f$$

- c) How many congruence classes are there in $\mathbb{Z}_5[x]/(x^3 + x + 1)$?

The number of congruence classes are the number of polynomials $f(x)$ such that $\deg(f(x)) < \deg(x^3 + x + 1)$,
So there are 5^3 congruence classes respectively for $(1, x, x^2)$



be a field and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F)$ such that $ac \neq 0$.
 Pf. ① we prove: if A is invertible then $ad - bc \neq 0_F$
 Assume $B = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$ is inverse of A so $\begin{pmatrix} x & y \\ z & w \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1_F & 0_F \\ 0_F & 1_F \end{pmatrix}$

$$\Rightarrow \begin{cases} ax + cy = 1_F \\ bx + dy = 0_F \\ az + cw = 0_F \\ bz + dw = 1_F \end{cases} \Rightarrow \begin{cases} abx + adx + bcy + cdw = 1_F & \text{①} \\ abx + adx + bcz + cdw = 0_F & \text{②} \end{cases}$$

 Assume $ad - bc = 0_F$, then $ad = bc$

$$\Rightarrow \text{①} = \text{②} \Rightarrow 0_F = 1_F \Rightarrow F = \{0_F\}$$

 which is impossible.

② We prove: if $ad-bc \neq 0$ then A is invertible.

Consider $B = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, so $AB = BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

So B is inverse of A . By (00), A is invertible
iff $ad - bc \neq 0_F$.

b) Prove that A in part (a) is a zero divisor if and only if $ad - bc = 0_F$.

Pf. Assume $ad - bc = 0_F$, then $A \cdot \begin{bmatrix} d & b \\ c & a \end{bmatrix} = \begin{bmatrix} 0_F & 0_F \\ 0_F & 0_F \end{bmatrix}$

$\widehat{= \begin{bmatrix} d & b \\ c & a \end{bmatrix} A}$

So A is a zero divisor

Therefore $ad - bc = 0 \Rightarrow A$ is zero divisor

Assume λ is a zero divisor

So $\exists B$ s.t. $BA = 0$ where $B \neq 0$

let $B = \begin{bmatrix} x & y \\ z & u \end{bmatrix} \Rightarrow \begin{cases} ax + cy = 0_i \\ bx + dy = 0_F \\ az + cu = 0_F \\ bz + du = 0_F \end{cases}$

Therefore $ad - bc = \Delta F$

$\Rightarrow A$ is zero divisor

1

0

Problem 5 Continued...

- c) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a matrix in $M_2(\mathbb{Z})$. Show that A is invertible in $M_2(\mathbb{Z})$ if $ad - bc = \pm 1$.

Pf Assume $ad - bc = \pm 1$

Consider $B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$\text{Then } AB = BA = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}$$

Therefore B is inverse of A

Therefore A is invertible in $M_2(\mathbb{Z})$ if $ad - bc = \pm 1$

- d) If $ad - bc \neq 0, 1, \text{ or } -1$, show that A is neither a unit nor a zero divisor in $M_2(\mathbb{Z})$.

Pf Assume for contradiction that A is a unit

so $\exists B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$ for some $x, y, z, w \in \mathbb{Z}$ s.t.

$$AB = BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



Problem 6 (16 points). There are four parts to this problem (see next page).

- a) Prove that the ideal $(x^2 + 3x, 3x + 2) \subset \mathbb{Z}_7[x]$ is principal by finding an element $c \in \mathbb{Z}_7[x]$ that generates the ideal.

Since $\mathbb{Z}_7$ is a field,

$$\{p(x^2 + 3x) + q(3x + 2) \mid p, q \in \mathbb{Z}_7[x]\} = \{k \gcd(x^2 + 3x, 3x + 2) \mid k \in \mathbb{Z}_7[x]\}$$

or we have proved.

$$\text{Since } x^2 + 3x = x(x + 3),$$

$$3x + 2 = 3(x + 3)$$

$$\gcd(x^2 + 3x, 3x + 2) = x + 3$$

$$\text{So } (x^2 + 3x, 3x + 2) = (x + 3)$$

- b) Prove that the ideal $(2, x) \subset \mathbb{Z}_8[x]$ is principal by finding an element $c \in \mathbb{Z}_8[x]$ that generates the ideal.

$$(2, x) = \{2p + qx \mid p, q \in \mathbb{Z}_8[x]\}$$

Problem 6 Continued...

- c) Let $D = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \mid a, b \in \mathbb{R} \right\} \subset M_2(\mathbb{R})$. Is D an ideal? Justify your answer.

It is ^{not} an ideal

Pf. ~~It is closed under addition~~

~~Select $A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, B = \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \in D$, $A+B = \begin{pmatrix} a+c & 0 \\ 0 & b+d \end{pmatrix} \in D$~~

Consider $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \in D$

and $C = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$

So $AC = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \notin D$

Therefore not an ideal.

- d) Let $\mathbb{F}$ be a field and let $I = (f(x))$ be an ideal in $\mathbb{F}[x]$. Suppose that $g(x) + I$ is a unit in $\mathbb{F}[x]/I$. Prove that the ideal $(f(x), g(x)) \subset \mathbb{F}[x]$ is all of $\mathbb{F}[x]$.

Pf Since $g(x) + I$ is a unit in $\mathbb{F}[x]/I$

$$\underline{g(x) \neq 0_{\mathbb{F}}} \text{ and } \exists h(x) \text{ s.t. } \underbrace{(h(x) + I)}_{\in \mathbb{F}[x]} (g(x) + I) = (1_{\mathbb{F}} + I)$$

$$\text{So } h(x)g(x) - 1_{\mathbb{F}} \in I$$

$$\text{So } h(x)g(x) - 1_{\mathbb{F}} = k(x)f(x) \text{ for some } k(x) \in \mathbb{F}[x]$$

$$\text{Therefore } 1_{\mathbb{F}} = h(x)g(x) - k(x)f(x)$$

$$\text{So } \underline{1_{\mathbb{F}} \in (f(x), g(x))}$$

Therefore $(f(x), g(x)) = \mathbb{F}[x]$ since $\forall q(x) \in \mathbb{F}[x]$

$$q(x) \cdot 1 = q(x) \in (f(x), g(x))$$



Problem 7 Continued...

c) Let I and J be ideals in a ring R . Prove that

$$\frac{I}{I \cap J} \cong \frac{I+J}{J}.$$

Consider the homomorphism

$$\varphi: I \rightarrow (I+J)/J$$

sending $\underbrace{a}_{+b} \mapsto \underbrace{a+J}_{b+J}$ where $a \in I$ and $b \in J$

$$\ker(\varphi) = I \cap J$$

By the first isomorphism theorem,

$$I/I \cap J \cong (I+J)/J$$



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